

A6 – Elastic Load Balancing

Group 1

1. Reuse the VPC configuration you created in your VPC assignments or use VPC Default

The screenshot shows the AWS VPC console interface. On the left, the 'VPC dashboard' sidebar lists various network resources. The main area displays 'Your VPCs (1/2)' with a table listing VPCs. The 'project-vpc' is selected, and its details are shown below the table.

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-0cc4c63a99ec930c0	Available	Off	172.31.0.0/16	-
project-vpc	vpc-0701ecdb9b6af1d4d	Available	Off	10.0.0.0/16	-

vpc-0701ecdb9b6af1d4d / project-vpc

Details

VPC ID vpc-0701ecdb9b6af1d4d	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-01bf5d42d77072a3c	Main route table rtb-0629ae93d4b0a316f
Main network ACL acl-0a31d8c5f6f6a3e3	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool

2. Launch 2 EC2s in 2 different AZs and install, start, and enable Apache
Creating an EC2 instance called as EC2-Server-1

Inbox (2,682) v.deshmukh@... Content Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances

Search [Alt+S] United States (N. Virginia) Account ID: 3866-9106-0433 Vajayanti-CC

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough Do not show me this message again.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

EC2-Server-1 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)

1

[Software Image \(AMI\)](#)
Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0bdd88bd06d16ba03

[Virtual server type \(instance type\)](#)
t3.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

VD-ELB [Create new key pair](#)

Network settings Info

VPC - *required* Info

vpc-0701ecdb9b6af1d4d (project-vpc)
10.0.0.0/16

Subnet Info

subnet-0733c170ea03d0c9c project-subnet-public1-us-east-1a
VPC: vpc-0701ecdb9b6af1d4d Owner: 386691060433
Availability Zone: us-east-1a (use 1-az1) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.0.0/20 [Create new subnet](#)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - *required*

ELB-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@#%&'()*~+-=;:[]{}|`~\$*

Description - *required* Info

Summary

Number of instances [Info](#)

1

[Software Image \(AMI\)](#)
Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0bdd88bd06d16ba03

[Virtual server type \(instance type\)](#)
t3.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

CloudShell Feedback Console Mobile App

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Creating SG with the necessary rules

Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search [Alt+S]

United States (N. Virginia) Account ID: 3866-9106-0433 Vajayanti-CC

EC2 > Instances > Launch an instance

Security group name - *required*

ELB-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@#%&*~

Description - *required* [Info](#)

Allows HTTP and SSH for ELB lab

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 161.253.2.253/32) [Remove](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#)

ssh TCP 22

Source type [Info](#) Name [Info](#) Description - *optional* [Info](#)

My IP Add CIDR, prefix list or security group e.g. SSH for admin desktop

161.253.2.253/32

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0) [Remove](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#)

HTTP TCP 80

Source type [Info](#) Source [Info](#) Description - *optional* [Info](#)

Anywhere Add CIDR, prefix list or security group e.g. SSH for admin desktop

0.0.0.0/0

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-0bdd88bd06d16ba03

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

In the advanced details section, in the "User data" box, I have pasted the following script so that it will automatically install and start Apache and create a test page.

Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search [Alt+S]

United States (N. Virginia) Account ID: 3866-9106-0433 Vajayanti-CC

EC2 > Instances > Launch an instance

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Select

User data - *optional* [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello from Server 1 (AZ-A)</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-0bdd88bd06d16ba03

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

Successfully launched EC2-Server-1 instance

Instance summary for i-084d447c36a12a5d4 (EC2-Server-1) Info

Updated less than a minute ago

Instance ID
i-084d447c36a12a5d4

Public IPv4 address
34.201.32.238 | [open address](#)

Private IPv4 addresses
10.0.1.181

Instance state
Running

Public DNS
ec2-34-201-32-238.compute-1.amazonaws.com | [open address](#)

Hostname type
IP name: ip-10-0-1-181.ec2.internal

Private IP DNS name (IPv4 only)
ip-10-0-1-181.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
34.201.32.238 [Public IP]

IAM Role
-

IMDSv2
Required

Operator
-

Instance type
t3.micro

VPC ID
vpc-0701ecd9b6af1d4d (project-vpc) | [open](#)

Subnet ID
subnet-0733c170ea03d0c9c (project-subnet-public1-us-east-1a) | [open](#)

Instance ARN
arn:aws:ec2:us-east-1:386691060433:instance/i-084d447c36a12a5d4

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendation | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Following the similar steps to create another EC2 instance in a different AZ

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name
EC2-Server-2 | [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances | Info
1

Software Image (AMI)
Amazon Linux 2023.9.2 AMI 2023.9.2...[read more](#)
ami-0bdd88bd06d16ba03

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Selecting a subnet that is in different AZ (us-east-1b)

Inbox (2,682) · v-deshmukh@g...Launch an instance | EC2 | us-east-1

Instance details | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Account ID: 3866-9106-0433Valjayanti-CCUnited States (N. Virginia)

Search[Alt+S]

EC2 > Instances > Launch an instance

▼ Network settings Info

VPC - required Info
vpc-0701ecdb9b6af1d4d (project-vpc)
10.0.0.0/16

Subnet Info
subnet-0bc9fc2596be46313 project-subnet-private2-us-east-1b
VPC: vpc-0701ecdb9b6af1d4d Owner: 386691060433
Availability Zone: us-east-1b (use1-aaz2) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.144.0/20

Create new subnet

Auto-assign public IP Info
Enable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups Info
Select security groups
ELB-SG sg-0383607ffafd37eb1 X
VPC: vpc-0701ecdb9b6af1d4d
Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-obdd88bd06d16ba03

Virtual server type (instance type)
t3.micro

Firewall (security group)
ELB-SG

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

▼ Configure storage Info

Configure storage

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same script, but changing the HTML line so that we know we're hitting the second server

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search

United States (N. Virginia)

Account ID: 3866-9106-0433

Vaijayanti-CC

EC2 > Instances > Launch an instance

Metadata response hop limit

2

Allow tags in metadata

Select

User data - optional

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello from Server 2 (AZ-B)</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more

ami-0bdd88bd06d16ba03

Virtual server type (instance type)

t3.micro

Firewall (security group)

ELB-SG

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Instance summary for i-0558abd9cb8ffdf73 (EC2-Server-2)'. The bottom screenshot shows the 'Instances (2)' list.

Instance summary for i-0558abd9cb8ffdf73 (EC2-Server-2)

Updated less than a minute ago

Instance ID
i-0558abd9cb8ffdf73

Public IPv4 address
44.203.195.26 | [open address](#)

Instance state
Running

Private IPv4 addresses
10.0.147.99

Public DNS
ec2-44-203-195-26.compute-1.amazonaws.com | [open address](#)

Private IP DNS name (IPv4 only)
ip-10-0-147-99.ec2.internal

Instance type
t3.micro

VPC ID
vpc-0701ecdb9b6af1d4d (project-vpc)

Subnet ID
subnet-0bc9fc2596be46313 (project-subnet-private2-us-east-1b)

Instance ARN
arn:aws:ec2:us-east-1:386691060433:instance/i-0558abd9cb8ffdf73

Auto-assigned IP address
44.203.195.26 [Public IP]

Hostname type
IP name: ip-10-0-147-99.ec2.internal

Answer private resource DNS name
-

IAM Role
-

IMDSv2
Required

Operator
-

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendation | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Instances (2)

Find Instance by attribute or tag (case-sensitive) | All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	EC2-Server-2	i-0558abd9cb8ffdf73	Running	t3.micro	3/3 checks pass	View alarms	us-east-1b	ec2-44-203
☐	EC2-Server-1	i-084d447c36a12a5d4	Running	t3.micro	3/3 checks pass	View alarms	us-east-1a	ec2-34-201

Select an instance

3. Create a Target Group and add these EC2s

- Step 1 Create target group
- Step 2 - recommended Register targets
- Step 3 Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

My-ELB-TG

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 9/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0701ecdb9b6af1d4d (project-vpc)

10.0.0.0/16



Create VPC

Protocol version

HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

Advanced health check settings

Attributes

EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Q Filter instances

< 1 >

☒	Instance ID	Name	State	Security groups	Zone
☒	i-0558abd9cb8ffdf73	EC2-Server-2	Running	ELB-SG	us-east-1b
☒	i-084d447c36a12a5d4	EC2-Server-1	Running	ELB-SG	us-east-1a

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (0)

Remove all pending

The screenshot shows the AWS Management Console interface for creating a target group. The top navigation bar includes the AWS logo, a search bar, and the current region (United States (N. Virginia)). The breadcrumb trail indicates the path: EC2 > Target groups > Create target group.

The main content area is titled "0 selected" and shows the "Ports for the selected instances" section. A text input field contains the value "80", and a note below it states "1-65535 (separate multiple ports with commas)". A button labeled "Include as pending below" is present.

Below this section, a message states: "2 selections are now pending below. Include more or register targets when ready."

The "Review targets" section displays a table of targets. The table has columns: Instance ID, Name, Port, State, Security groups, Zone, Private IPv4 address, and Subnet ID. Two targets are listed, both in a "Running" state.

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0558abd9cb8ffdf73	EC2-Server-2	80	Running	ELB-SG	us-east-1b	10.0.147.99	subnet-0bc9fc2596be46
i-084d447c36a12a5d4	EC2-Server-1	80	Running	ELB-SG	us-east-1a	10.0.1.181	subnet-0733c170ea03dc

At the bottom of the "Review targets" section, a message states: "2 pending". Navigation buttons include "Cancel", "Previous", and "Next".

Target Group created

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#TargetGroup:targetGroupArn=arn:aws:elasticloadbalancing:us-east-1:386691060433:targetgroup/My-ELB-TG/13...

EC2 > Target groups > My-ELB-TG

Capacity Manager New

Images

- AMIs
- AMI Catalog

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

Auto Scaling

- Auto Scaling Groups

Settings

Successfully created the target group: **My-ELB-TG**. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the **Targets** tab.

My-ELB-TG

Details

arn:aws:elasticloadbalancing:us-east-1:386691060433:targetgroup/My-ELB-TG/131f5d480c315f0c

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-0701ecdb9b6af1d4d
IP address type IPv4	Load balancer None associated		

2 Total targets	0 Healthy	0 Unhealthy	2 Unused	0 Initial	0 Draining
0 Anomalous					

► **Distribution of targets by Availability Zone (AZ)**
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

[Filter targets](#)

	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...
☐	i-0558abd9cb8ffdf73	EC2-Server-2	80	us-east-1b (us...	Unused	Target group is not co...	-	-
☐	i-084d447c36a12a5d4	EC2-Server-1	80	us-east-1a (us...	Unused	Target group is not co...	-	-

4. Create an ELB, i.e., MyElb, and make sure these 2 EC2s are part of the ELB target group
Creating an application LB

EC2 > Load balancers > Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

MyElb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with Internet-facing load balancers only.

Network mapping [Info](#)

EC2 > Load balancers > Create Application Load Balancer

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0733c170ea03d0c9c
IPv4 subnet CIDR: 10.0.0.0/20

project-subnet-public1-us-east-1a

☒ us-east-1b (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0bc9fc2596be46313
IPv4 subnet CIDR: 10.0.144.0/20

project-subnet-private2-us-east-1b

⚠ The selected subnet does not have a route to an internet gateway. This means that your load balancer will not receive internet traffic.
You can proceed with this selection; however, for internet traffic to reach your load balancer, you must update the subnet's route table in the [VPC console](#).

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

ELB-SG
sg-0383607ffafd37eb1 VPC: vpc-0701ecdb9b6af1d4d

The first screenshot shows the 'Create Application Load Balancer' wizard in the AWS Management Console. The 'Listeners and routing' section is active, showing a listener for HTTP on port 80. The default action is 'Forward to target groups', and the target group is 'My-ELB-TG' with a weight of 1 and 100% percent. The second screenshot shows the 'MyElb' details page, which includes information about the load balancer type (Application), status (Active), VPC (vpc-0701ecdb9b6af1d4d), availability zones (subnet-0733c170ea03d0c9c and subnet-0bc9fc2596be46313), and the DNS name (MyElb-619332181.us-east-1.elb.amazonaws.com).

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Protocol: HTTP Port: 80

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group

Choose a target group and specify routing weight or [create target group](#).

Target group

My-ELB-TG Target type: Instance, IPv4 | Target stickiness: Off HTTP Weight: 1 Percent: 100%

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness, the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

MyElb

Details

Load balancer type: Application Status: Active VPC: vpc-0701ecdb9b6af1d4d Load balancer IP address type: IPv4

Scheme: Internet-facing Hosted zone: Z35XD0TRQ7X7K Availability Zones: subnet-0733c170ea03d0c9c (us-east-1a (use1-az1)) subnet-0bc9fc2596be46313 (us-east-1b (use1-az2)) Date created: November 4, 2025, 14:42 (UTC-05:00)

Load balancer ARN: arn:aws:elasticloadbalancing:us-east-1:386691060433:loadbalancer/app/MyElb/785b1fbfadff516d DNS name: MyElb-619332181.us-east-1.elb.amazonaws.com (A Record)

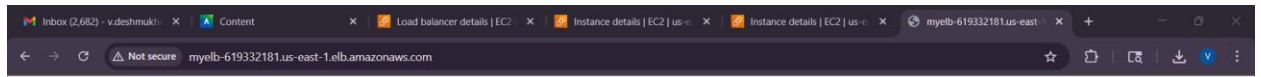
Listeners and rules (1)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

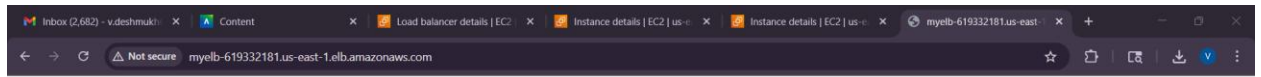
Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate
HTTP:80	Forward to target group My-ELB-TG (100%)	1 rule	ARN	Not applicable	Not applicable

- Access the ELB via the web
DNS Name : MyElb-619332181.us-east-1.elb.amazonaws.com



Hello from Server 1 (AZ-A)

Upon refreshing the page



Hello from Server 2 (AZ-B)

6. Stop an EC2 and access again the ELB via the web
Stopping EC2-Server-1 instance

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Instances' page for the 'us-east-1' region. It lists two EC2 instances: 'EC2-Server-2' (Running) and 'EC2-Server-1' (Stopped). The bottom screenshot shows the 'My-ELB-TG' target group details. It indicates that the target group has 2 total targets, with 1 healthy and 0 unhealthy. The 'Registered targets' table shows that 'EC2-Server-2' is healthy and 'EC2-Server-1' is unused.

Instances (2) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
EC2-Server-2	i-0558abd9cb8ffdf73	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	ec2-44-203
EC2-Server-1	i-084d447c36a12a5d4	Stopped	t3.micro	-	View alarms +	us-east-1a	-

Select an instance

Target groups > My-ELB-TG

Instance: HTTP: 80, HTTP1: vpc-0701ecdb9b6af1d4d

IP address type: IPv4, Load balancer: MyElb

2 Total targets: 1 Healthy, 0 Unhealthy, 0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

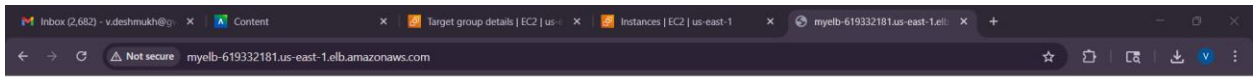
Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) Info

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

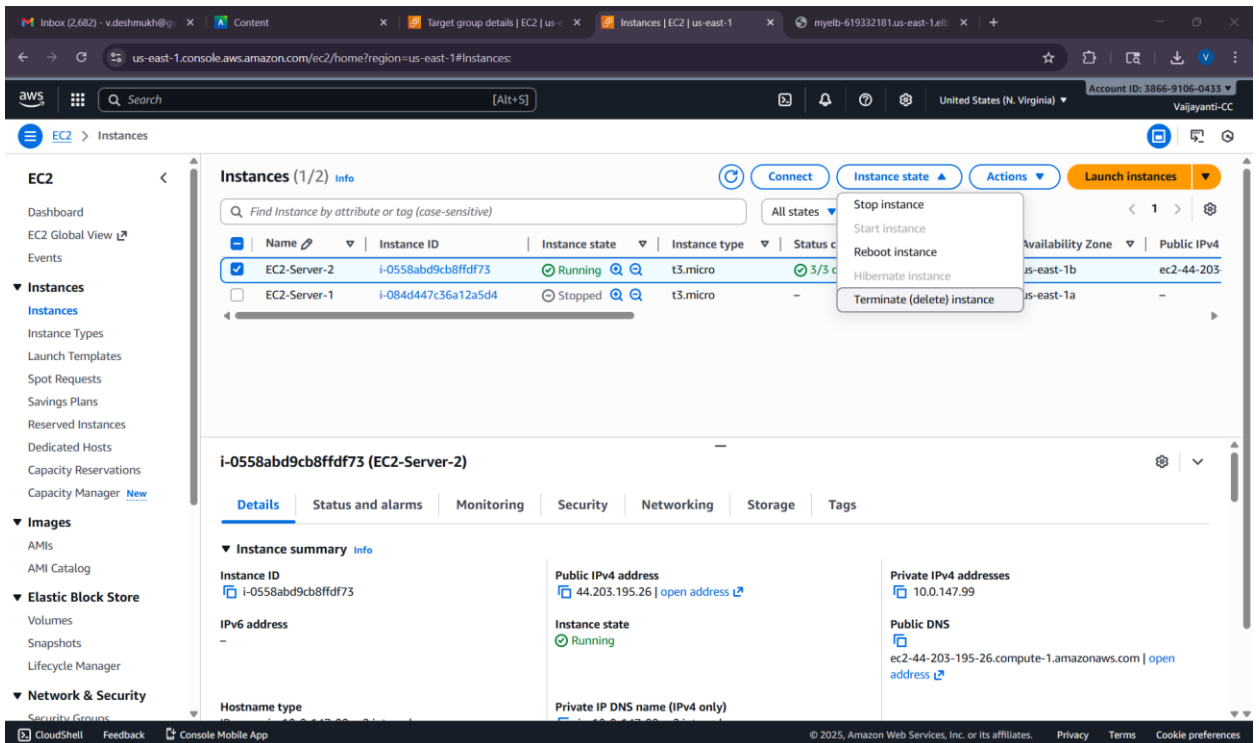
Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...
i-0558abd9cb8ffdf73	EC2-Server-2	80	us-east-1b (us...)	Healthy	-	No override.	No over
i-084d447c36a12a5d4	EC2-Server-1	80	us-east-1a (us...)	Unused	Target is in the stoppe...	No override.	No over

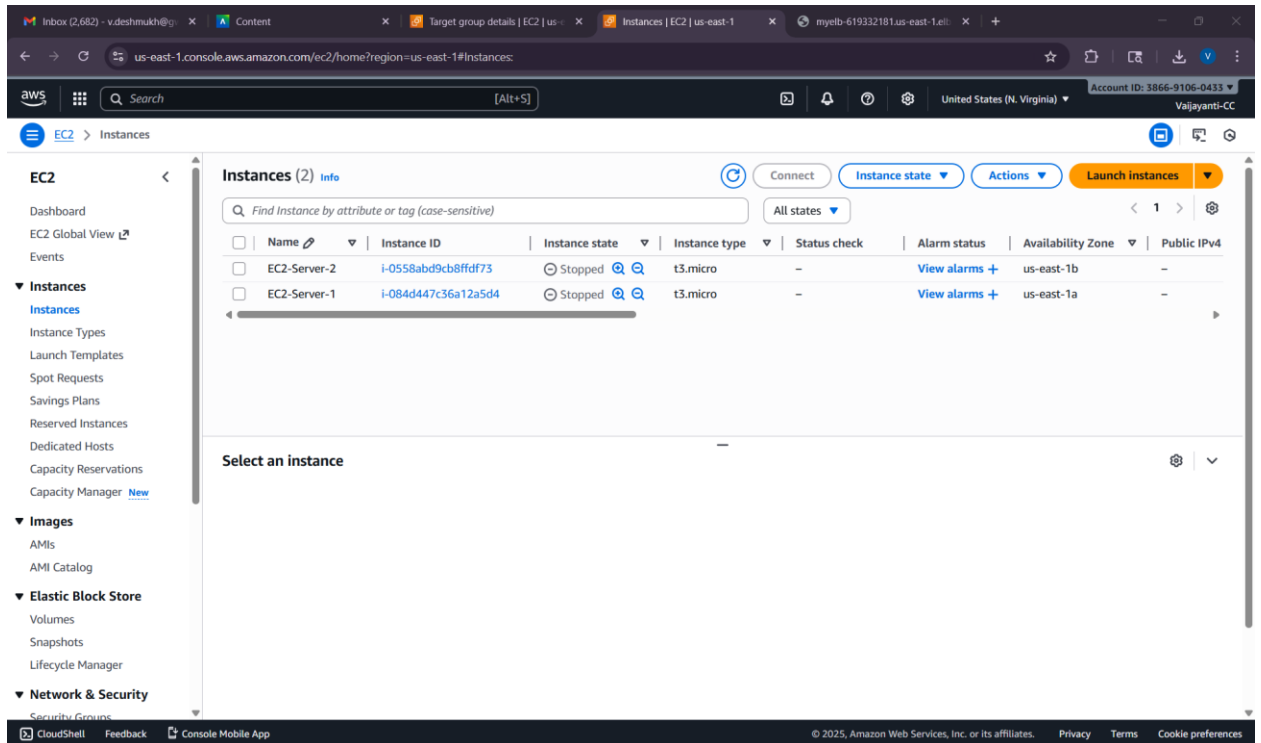
Only “Hello from Server 2 (AZ-B)” can be seen over and over. The load balancer knows Server 1 is stopped, so it automatically stops sending any traffic to it. The website stays online and is served by the other healthy instance.



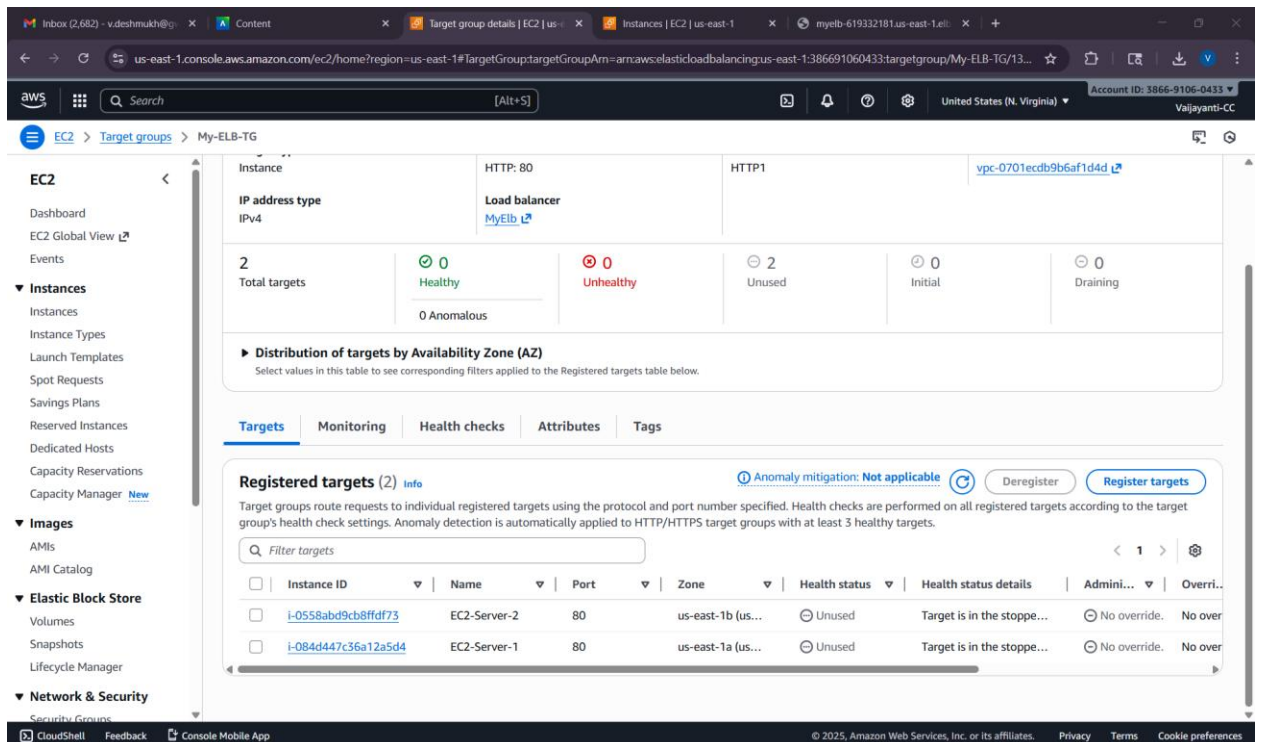
Hello from Server 2 (AZ-B)

7. Stop the 2nd EC2 and access again the ELB via the web
Stopping EC2-Server-2 instance

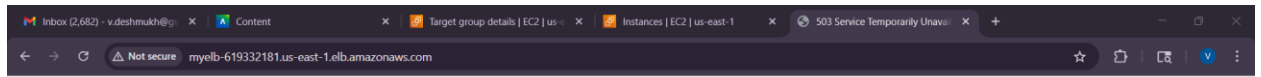




The status of EC2-Server-2 instance in TG changed from 'healthy' to 'unused'



Upon refreshing the website



503 Service Temporarily Unavailable

8. Describe results

The application load balancer successfully distributes the traffic across both ec2 instances. Upon stopping EC2-Server-1, the ELB health checks correctly identified the instance and all subsequent traffic was automatically rerouted to the remaining EC2-Server-2, which consistently displayed the Server 2 (AZ-B) page. This demonstrated the ELB's high availability by seamlessly handling a single-instance failure.

After stopping the second instance, EC2-Server-2, the ELB detected that no healthy targets remained in its target group. Hence when I tried accessing the dns, it failed (503 Service Unavailable)